

CODEGRAPH

CodeGraph — HelixCode Integration

Overview

CodeGraph (@colbymchenry/codegraph) is a local SQLite semantic code-knowledge-graph exposed to agents over MCP. It provides 100% local, zero-cloud code intelligence for all CLI agents working on HelixCode.

Constitution reference: §11.4.78, §11.4.79, §11.4.80

Installation

CodeGraph is installed globally:

```
npm install -g @colbymchenry/codegraph  
# Binary: /Users/milosvasic/.local/bin/codegraph
```

Configuration

IMPORTANT (verified 2026-06-22): CodeGraph v1.0.1 is **zero-config** and derives its exclusion scope **only from .gitignore** (+ .git/info/exclude), **NOT** from .codegraph/config.json. The legacy config.json in this repo is **inert** — it is no longer read by the tool. Editing it has no effect on what gets indexed. The single source of truth for exclusion is the ignore files. (Ref: <https://colbymchenry.github.io/codegraph/getting-started/configuration/>)

Exclusion source of truth: .gitignore (repo-tracked) and/or .git/info/exclude (local, non-tracking — preferred for paths that must stay git-tracked, e.g. cli_agents/).

Must be excluded (per §11.4.10 + §11.4.79) — verify each is actually absent from the index, NOT merely listed in the inert config: - cli_agents/, cli_agents_resources/, github_pages_website/ — third-party - nested submodules/**/submodules/**, submodules/**/cli_agents/** — vendored trees - generated **/*.gen.ts, **/worker-configuration.d.ts - **/.env, **/.env.*, **/*.key, **/*.pem, **/secrets/** — credentials

Must be included (own-org per §11.4.79): submodules/*, constitution/, helix_code/ — confirmed present via live codegraph_explore.

Anti-bluff note (§11.4): codegraph_validate.sh historically validated config.json's lists — which the tool ignores — so a 18/18 PASS did NOT prove third-party paths were excluded (36k cli_agents files were in fact indexed, 4.38 GB DB → #850 watchdog kills). The validator MUST instead query the live DB (SELECT COUNT(*) FROM files WHERE path LIKE '<excluded>/%' MUST be 0). See docs/research/codegraph_daemon_stability_20260622/findings.md.

Tool selection — codegraph_search vs codegraph_explore

codegraph_search is a SQLite FTS5 **implicit-AND symbol-NAME lookup** (matches a single symbol whose name/qualified-name/signature/docstring contains **all** the query tokens). A natural-language phrase like "provider Generate LLM" correctly returns **“No results found”** — no single symbol is named that. This is working-as-designed, **not** a broken index.

- **Name a symbol** → **codegraph_search** (e.g. OllamaProvider, NewOllamaProvider).
- **Ask a question / explore an area / multi-concept** → **codegraph_explore** (NL surface).

(Ref: docs/research/codegraph_search_rootcause_20260622/findings.md.)

Usage

Index the project

codegraph index

Check status

codegraph status

Start MCP server (for CLI agents)

codegraph serve --mcp --path /Volumes/T7/Projects/helix_code

MCP Integration

Configured in .mcp.json:

```
{
  "mcpServers": {
    "codegraph": {
      "type": "stdio",
      "command": "/Users/milosvasic/.local/bin/codegraph",
```

```

    "args": ["serve", "--mcp", "--path", "/Volumes/T7/Projects/
      helix_code"]
  }
}
}

```

Regeneration Mechanism (§11.4.77)

The `.codegraph/codegraph.db` file is gitignored. Regeneration:

```
codegraph index
```

Verification

```
codegraph status
```

Should show: indexed files, total symbols, database size

Cross-references

- constitution/Constitution.md §11.4.78 (CodeGraph mandate)
- constitution/Constitution.md §11.4.79 (own-org submodules included)
- constitution/Constitution.md §11.4.80 (regular update + sync)
- constitution/Constitution.md §11.4.77 (regeneration mechanism)

Sources verified 2026-06-22: <https://registry.npmjs.org/@colbymchenry/codegraph/latest> , <https://colbymchenry.github.io/codegraph/getting-started/configuration/>

Cross-referenced this doc against the latest official CodeGraph sources (fetched 2026-06-22). Findings: - **Version current.** npm `@colbymchenry/codegraph latest` = **1.0.1** (registry latest tag; description “Local-first code intelligence for AI agents (MCP). Self-contained — bundles its own runtime.”). This doc’s “CodeGraph v1.0.1” claim is accurate. - **Exclusion source-of-truth confirmed.** The official configuration page confirms CodeGraph is “zero-config” with “no config file to write or keep in sync” — exclusions are driven by `.gitignore` (honored via git in git repos, read directly root+nested in non-git projects) plus built-in defaults (`node_modules`, `vendor`, `dist`, `build`, `target`, `.venv`, `Pods`, `.next`, files > 1 MB), with !negation overrides in `.gitignore`. This corroborates the doc’s “config.json is inert; `.gitignore/.git/info/exclude` are the single source of truth” claim — NO contradiction. - **Negative finding.** The fetched configuration page does not document a `codegraph sync` subcommand under that exact name; the §11.4.80 sync flow this doc references comes from the constitution-submodule `codegraph_*.sh` wrappers, not the upstream CLI surface. Treat the upstream CLI contract as authoritative for raw codegraph invocations.